

# The trinity force

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## Abstract

In this article, we present the proofs of three theorems related to the foundations of Functional Analysis. The first one deals with the question of whether, given a linear functional defined on a subspace can be extended to the entire space without increasing its norm. the second one states if the operators are bounded point by point, then their norms are uniformly bounded and finally if we have a continuous linear operator and surjective between Banach spaces preserve

## 1 Introduction

Despite that Stefan Banach no study mathematic formally yours apports was a big change in the era even more after he met Hugo Steinhaus in Kraków-Poland in 1916 where Banach begins to publish articles